Testbook: Leveling the playing field for millions of students in Tier 3 and 4 cities with technology

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Testbook used G
Cloud to create a
scalable and low-
platform to help
underprivileged s
succeed in public
placement exam:
prepare for a priv
career.

About Testbook

Testbook is the fastest-rising startup in India's edtech space. It is the go-to destination for over 3.2 million students to prepare for over 900 placement exams for India's public service sector, many of whom come from Tier 3 and Tier 4 cities where high-quality preparation resources are out of reach.

Industries: Technology

Google Cloud results

- Creates a highly scalable and low-cost platform for 3.2 million users
- Serves over 2.5 billion mock exam tries across 900 different exam types
- Creates a question difficulty scoring algorithm in 2 weeks
- Helps over 60,000 Indian

Empow underp student attemp exams

Location: India

Dataflow (https://cloud.google.com/dataflow/),

About Searce

Searce is a cloud consulting business that specializes in cloud data engineering, AI and machine learning, as well as advanced cloud infrastructure technology, such as Anthos and Kubernetes. The business has moved 3,000+ clients successfully to the cloud, and is part of the Google Cloud partner network.

- Google Cloud (https://cloud.google.com/)
- BigQuery (https://cloud.google.com/bigquery/)
- Cloud CDN (https://cloud.google.com/cdn/)
- Logging (https://cloud.google.com/logging/)
- Cloud Storage (https://cloud.google.com/storage/)
- Compute Engine (https://cloud.google.com/compute/)
- Dataflow (https://cloud.google.com/dataflow/)
- Cloud Dataproc (https://cloud.google.com/dataproc/)
- Firebase (https://firebase.google.com/)

students get public-sector jobs	Cloud Pub/Sub (https://cloud.google.com/pubsub/) Google Workspace (https://workspace.google.com/) Cloud Translation API (https://cloud.google.com/translate/)

Students in India's Tier 3 and Tier 4 cities often need better teaching resources to prepare for some of the most important tests of their lives. These tests grant tertiary students aged 18 to 30 placements in desirable public service jobs that can prove to be life-changing.

The competition is stiff, with many applicants vying for relatively few jobs. That is why Testbook (https://testbook.com/), India's leading exam preparation destination, is trying to help level the playing field for millions of hopeful Indian students.

"We started Testbook in 2014 in response to a major shift in the Indian exam ecosystem that saw a transition from paper-and-pen to computer-based tests," explains Ayush Varshney, the Chief Technology Officer of Testbook.

"We built a mock exam platform for these underprivileged students to practice and learn using high-quality content. The response has been very strong, and we see students wanting to continually practice and spend a lot of time on the platform, thus driving engagement up," he adds.

Today, Testbook offers preparation courses for 900 exams and has seen its rapid rise in tandem with the drop in 4G data plan prices for mobile devices. However, it takes a smart tech stack to help them ensure service delivery while keeping costs low for students.

Scaling service delivery during seasonal surges

When exam preparations depend entirely on unpredictable announcements of exam commencement, Testbook must always be ready to handle a massive traffic surge. As a result, they decided on a cloud-first strategy with Google Cloud (https://cloud.google.com/) since Testbook's inception to meet massive seasonality demand with a small team.

"When these exams are announced, we must be prepared to handle a 100 to 200 times spike in traffic," says Varshney.

"We did many proofs of concepts to determine the kind of throughput that we can serve with Managed Instance Groups. We discovered that using this solution would be very cost-effective because our back end is written in Golang, which has high direct kernel efficiency."

—Ayush Varshney, Chief Technology Officer, Testbook

In response to this challenge, the team turned to Google Cloud's Managed Instance Groups (https://cloud.google.com/compute/docs/instance-groups) to easily and quickly scale without having to write their code for deployment or needing deep knowledge of the solution at the start. They routinely run 150 VM instances daily, which can scale to 2,000 instances when exams are announced.

"We did many proofs of concepts to determine the kind of throughput that we can serve with Managed Instance Groups. We discovered that using this solution would be very cost-effective because our back end is written in Golang, which has high direct kernel efficiency," Varshney explains.

Delivering more value to stakeholders while keeping costs low

His team uses BigQuery (https://cloud.google.com/bigquery) as the platform's main data lake. They rely on its easy integration with other Google solutions like Google Analytics (https://marketingplatform.google.com/about/analytics/), Firebase (https://firebase.google.com/), and YouTube (https://www.youtube.com/) that can easily export data to BigQuery. They also self-deployed Pub/Sub (https://cloud.google.com/pubsub) to queue events on high-traffic days so that their infrastructure will not break down with the surge in traffic.

"The fact that we're charged only for what we query helps keep costs low. We also get to deepen our business intelligence capabilities, enabling our teams to be highly data-driven. BigQuery has been a big help for us, and if we were to do that internally or use another vendor, it would be prohibitively expensive," Varshney confides.

To build its question difficulty algorithm, the team uses fully managed services by Dataproc (https://cloud.google.com/dataproc) to make sense of the raw data generated by its 3.2 million student users. "If we had to build the algorithm from scratch, it would take months. But we did it in Dataproc, which took a couple of weeks," says Varshney. Meanwhile, Dataflow (https://cloud.google.com/dataflow) enables some of these analytics requirements to help Testbook identify what users search on the platform and what they click on.

Given the language diversity in India, Testbook also needs to deliver these exams in 14 different languages. To accomplish this, the team uses Translation AI (https://cloud.google.com/translate) to produce a draft version of the exam in each of the 14 languages, and then gets an internal reviewer or translator to ensure it is ready for use. It also translates all of Testbook's content, questions, and study notes.

Furthermore, the highly managed nature of Google Cloud services means low maintenance and monitoring needs for Testbook's DevOps team, which currently consists of just three people.

"It also really helps that Google Cloud's services billing is transparent. There are no hidden layers in billing that we have to figure out, so we can easily plan for future use cases because we know that this is how much we are getting billed for a certain service. We can also rely on our partner, Searce (https://www.searce.com/), to solve our billing challenges, and learn how to build something on Google Cloud. This gives us the confidence to scale while keeping costs intact," Varshney says.

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>
> *—Ayush Varshney, Chief Technology Officer, Testbook*

Helping students and teachers do their best work

Within this integrated environment, Testbook can help students keep track of their progress and goals. One major way analytics is used to help them do their best work is by identifying their weak and strong subjects, then suggesting the exam questions they should improve on.

"Each question is tagged with a difficulty value for every exam, and every time a user attempts a test, their competency scores will change in response to their state of preparedness," Varshney explains.

This analytics process happens throughout the three to four years a student spends preparing for these important exams, backed by an algorithm that considers their past and present performance data. It has also constantly been working across the 2.5 billion mock tests Testbook has served since its inception.

Furthermore, teachers need to know whether the mock tests that they create meet the needs of students, which is highly subjective. However, Testbook can quantify it using likes and view count data from YouTube to help teachers gauge the quality of their work.

"All in all, I think the high uptime and mock test quality are big reasons why there's a lot of trust in the Testbook platform, and why users enjoy using our platform. Our app is consistently rated at 4.6 stars on Google Play (https://play.google.com/store/apps?pli=1), and we feel that's a good measure of how happy our users are," says Varshney.

Growing from testing to upskilling

While Testbook has successfully placed 60,000 students into public sector jobs, the fact remains that many will not be able to secure one of the few jobs on offer. As a result, Testbook is looking to create a Skill Academy to help students prepare for the much more abundant private sector jobs in India.

"Our mission from the start has been to empower students from Tier 3 and Tier 4 cities whose parents do not have sufficient money to send them to big cities. The next phase of our platform is to turn it into a one-stop job preparation destination for the private sector," states Varshney.

"We will keep our platform highly affordable, easily accessible, and high-quality. Google Cloud has helped a startup like ours to scale while keeping costs low, and it will take us to our next phase with Skill Academy."

—Ayush Varshney, Chief Technology Officer, Testbook

This will mean even more scale is required for Testbook, both in terms of users and traffic. Initial trials have shown that Skill Academy graduates have been able to secure good digital jobs, even if they barely used a computer before.

"We will keep our platform highly affordable, easily accessible, and high-quality. Google Cloud has helped a startup like ours to scale while keeping costs low, and it will take us to our next phase with Skill Academy," Varshney concludes.